

Ex. No 9: Using Procmon to Identify Suspicious Processes

Date:

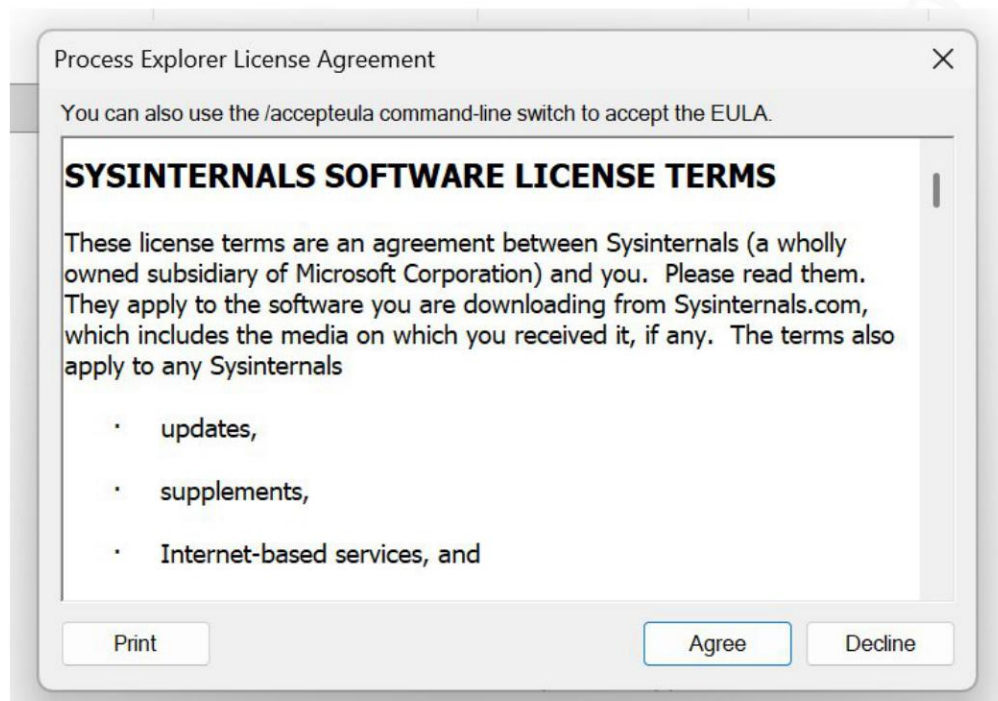
Aim

To monitor and identify suspicious process activities in a Windows system using Procmon (Process Monitor).

Procedure

1. Download and Extract Procmon

Obtain Process Monitor from the official Microsoft Sysinternals website and extract the downloaded ZIP file.



2. Run Procmon as Administrator

Right-click Procmon.exe → Run as Administrator to grant kernel-level system tracing access.

The screenshot displays the Windows Process Monitor (ProcMon) tool, which is monitoring file system activity. The interface includes a menu bar (File, Edit, Event, Filter, Tools, Options, Help) and a toolbar with various icons for filtering and viewing data. The main window is divided into several columns: Time, Process Name, PID, Operation, Path, Result, and Detail. The data shows a series of file reads and queries performed by Explorer.EXE, svchost.exe, lsass.exe, and ServiceHost.exe. The paths include system files like C:\Windows\System32\Windows.State..., C:\Windows\System32\StateRepository..., and C:\Windows\System32\lsassrv.dll, as well as user-specific paths like C:\Users\venkat\downloads\ProcessMo... and C:\Program Files\McAfee\WebAdvisor\... The results are all SUCCESS, and the detail column provides additional information such as offsets and lengths.

Time ...	Process Name	PID	Operation	Path	Result	Detail
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 86016, Len...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 69632, Len...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\MmCoreR.dll	SUCCESS	Offset: 1097728, L...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 786432, Le...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\MmCoreR.dll	SUCCESS	Offset: 1081344, L...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 778240, Le...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 733184, Le...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\MmCoreR.dll	SUCCESS	Offset: 1064960, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1675264, L...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\MmCoreR.dll	SUCCESS	Offset: 1060864, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1687552, L...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 5156864, L...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\MmCoreR.dll	SUCCESS	Offset: 995328, Le...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1658880, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1667072, L...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 5115904, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1650688, L...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\windows.storag...	SUCCESS	Offset: 8376320, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1654784, L...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 5087232, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1552384, L...
14:17:...	Explorer.EXE	17608	ReadFile	C:\Windows\System32\windows.storag...	SUCCESS	Offset: 8065024, L...
14:17:...	lsass.exe	1172	ReadFile	C:\Windows\System32\lsassrv.dll	SUCCESS	Offset: 1552384, L...
14:17:...	svchost.exe	4136	ReadFile	C:\Windows\System32\Windows.State...	SUCCESS	Offset: 5070848, L...
14:17:...	lsass.exe	1172	QueryNameInfo...	C:\Users\venkat\downloads\ProcessMo...	SUCCESS	Name: 'venkat' 'ee...
14:17:...	lsass.exe	1172	QueryNameInfo...	C:\Users\venkat\downloads\ProcessMo...	SUCCESS	Name: venkat ee...
14:17:...	ServiceHost.exe	16504	ReadFile	C:\Program Files\McAfee\WebAdvisor\...	SUCCESS	Offset: 6288384, L...
14:17:...	ServiceHost.exe	16504	ReadFile	C:\Program Files\McAfee\WebAdvisor\...	SUCCESS	Offset: 6288384, L...

3. Start Capture

Click the Capture Events (Ctrl + E) icon to begin real-time event telemetry monitoring.

4. Perform System Activity

Carry out normal or suspicious actions, such as executing an unknown binary or modifying registry keys.

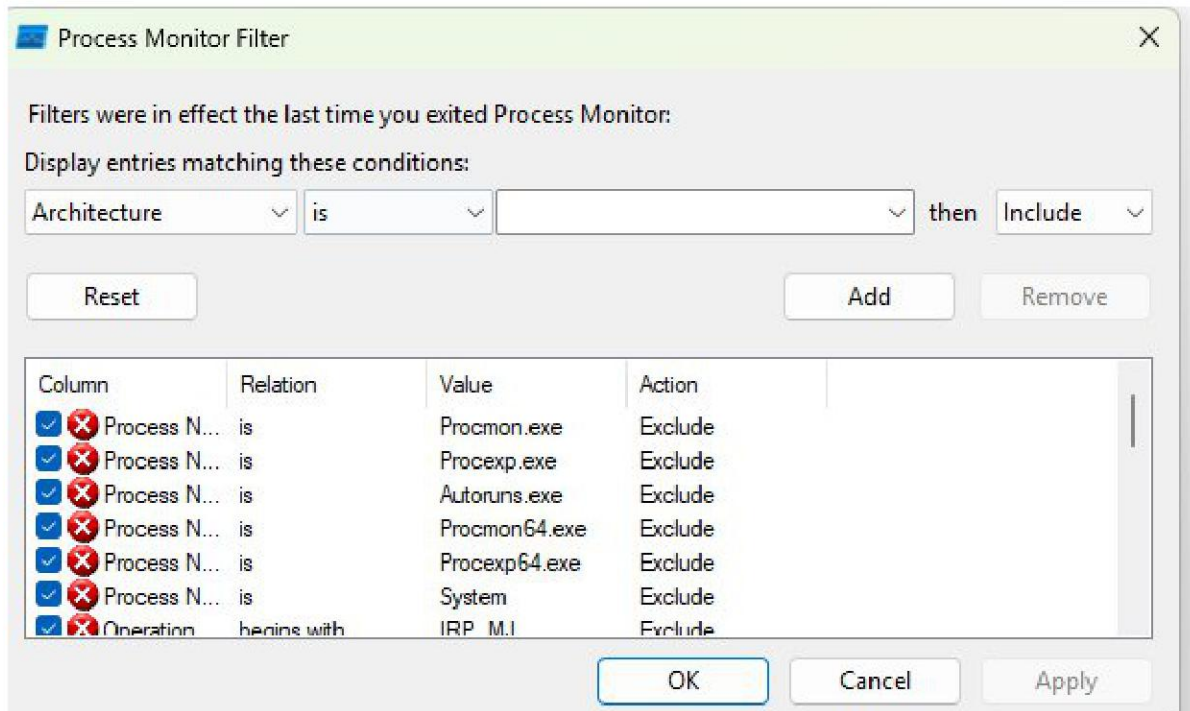
5. Stop Capture

Click the Capture icon again or press Ctrl + E once enough data has been collected.

6. Apply Forensic Filters

Go to Filter → Filter... (Ctrl + L) and configure relevant filter rules:

- Process Name contains suspicious.exe
- Operation contains WriteFile or CreateProcess
- Result is ACCESS DENIED or NAME NOT FOUND



7. Analyze Captured Events & Hierarchy

- **Process Tree:** Examine parent-child relationships in the Process Tree (Ctrl + T) to identify rogue spawned processes.
- **Activity Analysis:** Look for abnormal file writes, dropped binaries, and registry persistence keys.
- **Export:** Export relevant events as a .CSV file for investigative reporting.

8. Remediation & Threat Mitigation

Run a comprehensive antivirus scan and use malware removal tools for thorough cleanup. Verify executable signatures and review process file paths to ensure system integrity.

Rubrics

Criteria	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

- Thus, the analysis of suspicious processes using Process Monitor (Procmon) was performed successfully.
- All system and user processes were monitored with their respective file, registry, and process activities.
- Unsigned or abnormal processes were identified by verifying signatures, hierarchy trees, and file paths.